1.5 Graphs of Sine and Cosine Functions

Name _____ Period ____ Date _____

Pre-Calculus 1.5 (30-minute plan) · Student sheet

I can…

- find the amplitude and period of a sine or cosine function.
- describe phase shifts and vertical shifts.
- sketch a sine or cosine graph and write its equation.

Vocabulary · fill in the missing word

Amplitude — |a|, half the _____ between the maximum and minimum of y = a sin bx

Period — the _____ of one cycle: 2π/b for y = a sin bx

Phase shift — the _____ shift $\frac{c}{b}$ of y = a sin(bx − c)

Part 1 · Notes and examples (fill in each note, then work the example)

y = a sin(bx − c) + d: amplitude = |a|, period = _____.

Example 1

Find the amplitude and period.

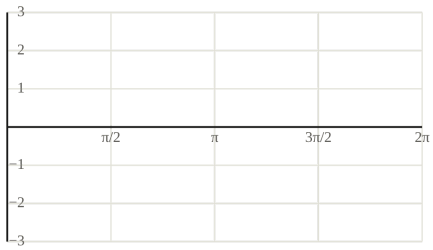
a. $y = 3 \sin 2x$

b. $y = -2 \cos(\frac{x}{2})$

Key points: one cycle splits into _____ equal parts: zero, max, zero, min, zero (for sine).

Example 2

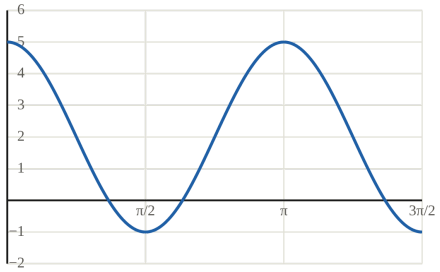
Find the five key points of y = 2 sin x on [0, 2π], then sketch.



Phase shift: one cycle starts where bx − c = 0 and ends where bx − c = _____.

Example 3

Describe y = 2 + 3 cos 2x: amplitude, period, maximum, and minimum.



Part 2 · Partner practice

Partner practice

Find one cycle of $y = \sin(2x - \pi)$.